

POLITECNICO DI MILANO



COMPUTATIONAL FINANCE

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Option Calibration and Pricing (HJM multifactor)

Group 11

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2 MODEL PARAMETERS ADMISSIBLE RANGE

Considering an Heath–Jarrow–Morton model, the general formula of the risk-neutral price dynamics of the swap is:

$$F(t, \tau_1, \tau_2) = F(0, \tau_1, \tau_2) \cdot \exp \left(\int_0^t A(u, \tau_1, \tau_2) du + \sum_{k=1}^p \int_0^t \Sigma_k(u, \tau_1, \tau_2) dW_k(u) + \sum_{j=1}^n \int_0^t \Upsilon_j(u, \tau_1, \tau_2) dJ_j(u) \right)$$

In our analysis, we use the HJM model for the 4Q23 German power swap driven by two Brownian motions, so $p=2$ and $n=0$, then our parameters are A , Σ_1 and Σ_2 .

These three parameters are real-valued continuous functions where $0 \leq t \leq \tau_1 \leq \tau_2$. It is also supposed that the two functions Σ_1 and Σ_2 are positive, so when considering Σ_1 and Σ_2 constant we have the only constraints $\Sigma_1 \geq 0$ and $\Sigma_2 \geq 0$.

3 MARTINGALE CONDITION

Since we need the explicit dynamics of the forward to be set under the risk-neutral probability Q , the price has to be a martingale. This entails a condition on A , that is, in the general case:

$$\begin{aligned} & \int_0^t A(u, \tau_1, \tau_2) du + \frac{1}{2} \sum_{k=1}^p \Sigma_k^2(u, \tau_1, \tau_2) + \sum_{j=1}^n \int_0^t \Upsilon_j(u, \tau_1, \tau_2) d\gamma_j(u) \\ & + \int_0^t \int_{\mathbb{R}} \left(e^{\Upsilon_j(u, \tau_1, \tau_2)z} - 1 - \Upsilon_j(u, \tau_1, \tau_2)z \mathbb{1}_{|z| < 1} \right) \nu_j(dz, du) = 0 \end{aligned}$$

With our conditions, this formula becomes:

$$\int_0^t A(u, \tau_1, \tau_2) du = -\frac{1}{2} \sum_{k=1}^p \Sigma_k^2(u, \tau_1, \tau_2)$$

which is further simplified to:

$$\int_0^t A(u, \tau_1, \tau_2) du = -\frac{1}{2} (\Sigma_1^2 + \Sigma_2^2).$$

4 CALIBRATION WITH BOTH PARAMETERS CONSTANT

We have to calibrate the model on the 4Q23 German option prices (the entire surface) by minimizing the distance between model and market prices.

In this point we consider both Σ_1 and Σ_2 constant.

First of all, we compute the market prices of the German option on the 4Q23, using the Black & Scholes formula starting from the given volatility matrix (TxN matrix of volatilities, one for each of the T=8 tenors and the N=21 strikes). In order to perform this computation we implement a function which exploits the MATLAB function *blkprice*, which uses Black's model for pricing futures options: European put and call. We take as F_0 the price of the future at *4-Nov-2022*. As time to expiration of the option we consider the tenors, while for the annualized futures price volatility we take the respective row from the volatility matrix for each tenor.

Then, we compute the model price exploiting the following formula to price a European call written on a swap contract:

$$C(t, T, K, \tau_1, \tau_2) = e^{-r(T-t)} F(t, \tau_1, \tau_2) \phi(d_1) - K \phi(d_2)$$

where:

$$d_1 = d_2 + \sqrt{(\Sigma_1^2 + \Sigma_2^2)t}$$

$$d2 = \frac{\log(F(t, \tau_1, \tau_2)/K) - 0.5(\Sigma_1^2 + \Sigma_2^2)t}{\sqrt{(\Sigma_1^2 + \Sigma_2^2)t}}$$

Eventually, we minimize the distance between market prices and model prices:

$$\min_{\Sigma_1, \Sigma_2} \sum |modelprice - marketprice|^2$$

with different methods implemented in MATLAB, such as *fmincon*, *lsqnonlin* and *fminsearch*, imposing the constraints of $\Sigma_1 \geq 0, \Sigma_2 \geq 0$.

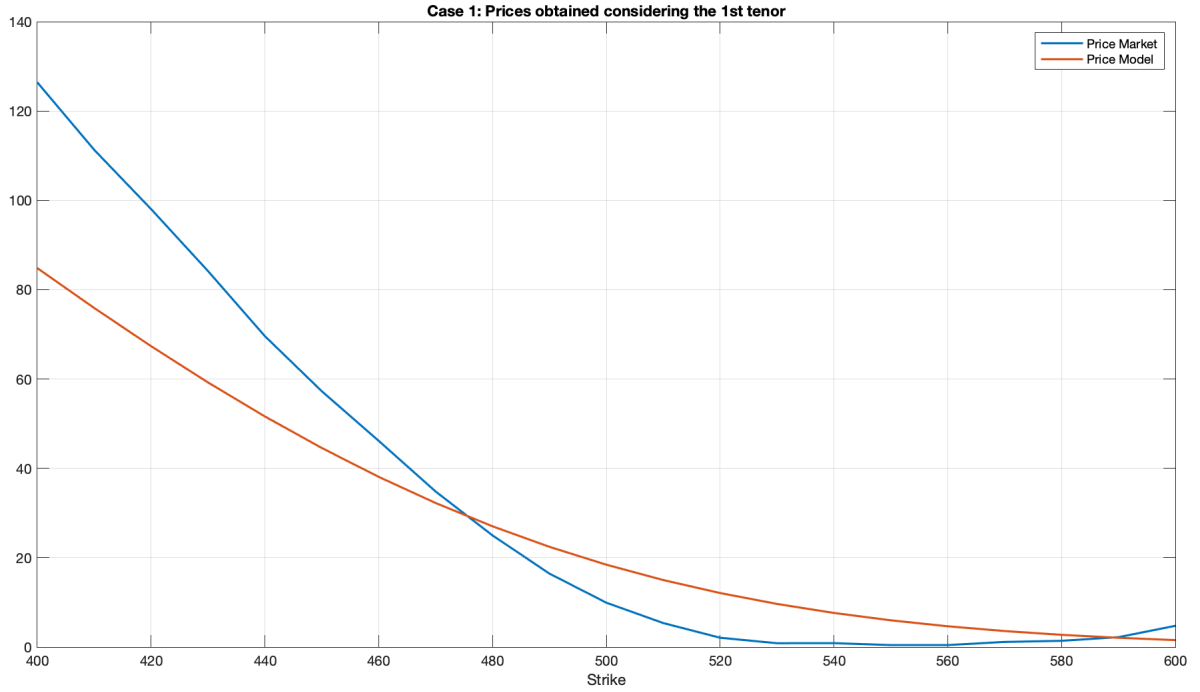


Figure 1: Comparison between Price Model and Price Market

We also compute the mean root square error through the formula:

$$MRSE = \sqrt{\frac{optimaldistance}{T \cdot N}},$$

in order to understand how good is our calibration.

The obtained values are:

	Σ_1	Σ_2	MRSE
<i>fmincon</i>	0.4193	0.4240	36.6742
<i>lsqnonlin</i>	0.2667	0.5334	36.6742
<i>fminsearch</i>	0.3142	0.5068	36.6742

Table 1: calibrated values of Σ_1 and Σ_2 and MRSE

Our results are acceptable because the MRSE has the same order of magnitude of the market prices. In addition, the sum of Σ_1 and Σ_2 in all the three cases is almost equal, which leads to the same MRSE.

Furthermore, we have to remember that the HJM framework is not free of disadvantages, in fact can be complicated to define a suitable volatility for practical purpose and it does not provide exact solution for pricing derivatives. In conclusion, we are quite satisfied with the results we have obtained, remembering that we are using an HJM simplified model with only two Brownian motions and no Π processes.

5 CALIBRATION WITH Σ_1 TIME DEPENDENT AND Σ_2 CONSTANT

Now we consider extending our model relaxing the hypothesis of Σ_1 constant, indeed $\Sigma_1 = \Sigma_1(t)$ will depend on time, while we assume Σ_2 to be constant and equal to 0.

The market prices remain unchanged, so we do not recompute them.

For the model prices instead we use the following formulas for the computation of d_1 and d_2 :

$$d_1 = d_2 + \sqrt{\int_0^t \Sigma_1^2(s, \tau_1, \tau_2) ds}$$

$$d_2 = \frac{\log(F(t, \tau_1, \tau_2)/K) - 0.5 \int_0^t \Sigma_1^2(s, \tau_1, \tau_2) ds}{\sqrt{\int_0^t \Sigma_1^2(s, \tau_1, \tau_2) ds}}$$

We minimize as before the distance between market prices and model prices, calibrating directly the integral of Σ_1^2 for each one of the 8 tenors we have, using the MATLAB methods *fmincon* and *lsqnonlin*.

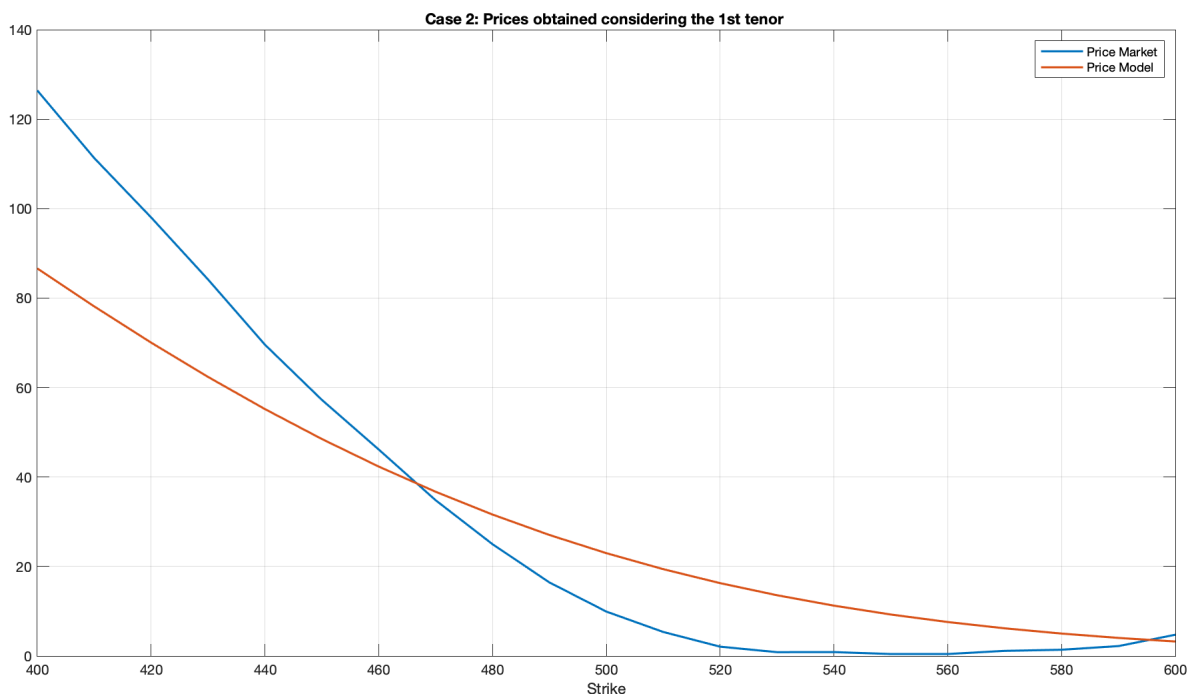


Figure 2: Comparison between Price Model and Price Market

The optimal values of $\int_0^t \Sigma_1^2 ds$ obtained for the 8 different tenors are:

Tenor	fmincon	lsqnonlin
0.05	0.0248	0.0248
0.10	0.0396	0.0396
0.15	0.0588	0.0588
0.20	0.0762	0.0762
0.25	0.0927	0.0927
0.30	0.1115	0.1115
0.40	0.1346	0.1346
0.50	0.1573	0.1573

Table 2: calibrated values of $\int_0^t \Sigma_1^2 ds$

As expected, we notice that they are increasing since the integrand is positive and the integration domain is increasing. The integral of Σ_1^2 for t that goes from 0 to the time to maturity is the only relevant one because it is the largest and it represents exactly the time horizon from the start date *4-Nov-2022* to the last tenor (0.5 years later). In addition we notice that both methods lead to the same results, which supports our analysis even more.

The mean root square error $\sqrt{\text{optimaldistance}/(TN)}$ is computed using the last integral, according to what just stated above and the results we obtain are:

	fmincon	lsqnonlin
MRSE	57.2535	57.2532

Table 3: MRSE when Σ_1 time dependent

This is completely acceptable since it has the same order of magnitude of the market prices and also because we are using a model with only Brownian motions and no II processes.

6 CALIBRATION WITH BOTH PARAMETERS TIME DEPENDENT

Now we consider further extending our model relaxing also the hypothesis of Σ_2 constant, indeed both $\Sigma_1 = \Sigma_1(t)$ and $\Sigma_2 = \Sigma_2(t)$ will depend on time.

The market prices remain unchanged, so we do not recompute them.

For the model prices instead we use the following formulas for d_1 and d_2 :

$$d1 = d2 + \sqrt{\sum_{k=1}^p \int_0^t \Sigma_k^2(s, \tau_1, \tau_2) ds}$$

$$d2 = \frac{\log(F(t, \tau_1, \tau_2)/K) - 0.5 \sum_{k=1}^p \int_0^t \Sigma_k^2(s, \tau_1, \tau_2) ds}{\sqrt{\sum_{k=1}^p \int_0^t \Sigma_k^2(s, \tau_1, \tau_2) ds}}$$

As before, we minimize the distance between market prices and model prices, calibrating directly the integrals of Σ_1^2 and Σ_2^2 , for each one of the 8 tenors we have, using the MATLAB methods *fmincon* and *lsqnonlin*.

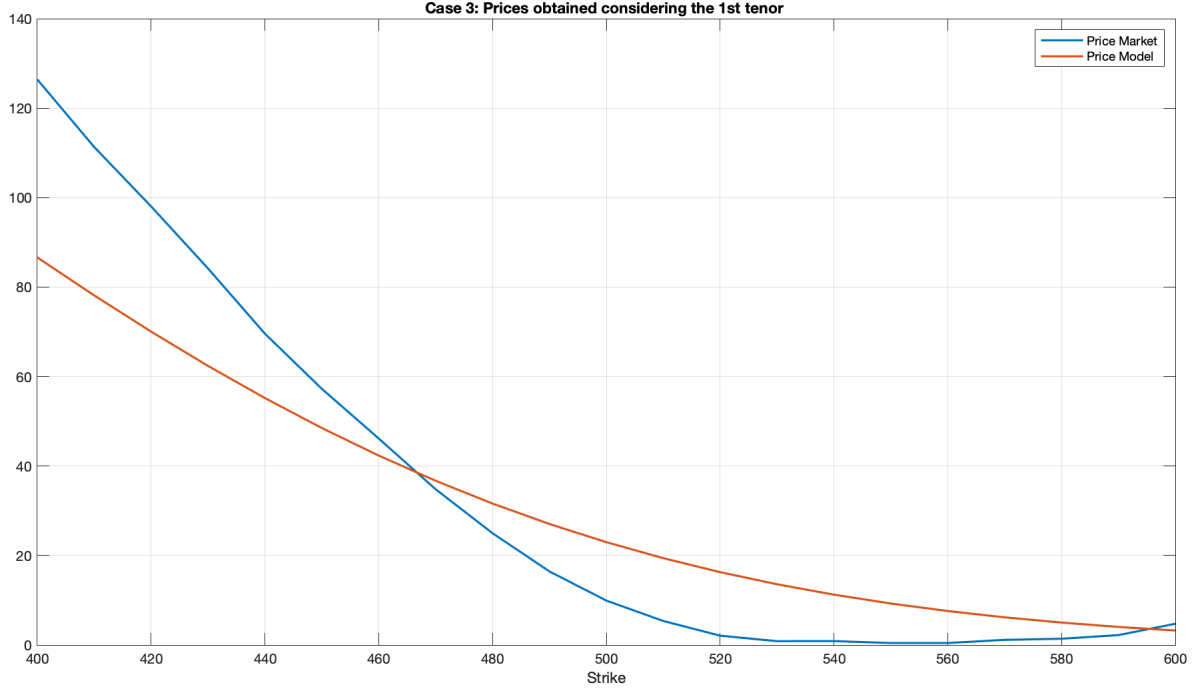


Figure 3: Comparison between Price Model and Price Market

The optimal values obtained are:

tenor	fmincon		lsqnonlin	
	$\int_0^t \Sigma_1^2 ds$	$\int_0^t \Sigma_2^2 ds$	$\int_0^t \Sigma_1^2 ds$	$\int_0^t \Sigma_2^2 ds$
0.05	0.0124	0.0124	0.0083	0.0165
0.10	0.0198	0.0198	0.0132	0.0264
0.15	0.0294	0.0294	0.0196	0.0392
0.20	0.0381	0.0381	0.0254	0.0508
0.25	0.0464	0.0464	0.0309	0.0618
0.30	0.0558	0.0558	0.0372	0.0744
0.40	0.0673	0.0673	0.0449	0.0897
0.50	0.0787	0.0787	0.0524	0.1049

Table 4: calibrated values of $\int_0^t \Sigma_1^2 ds$ and $\int_0^t \Sigma_2^2 ds$

As expected, we notice that they are increasing since the integrand is positive and the integration domain is increasing. In addition, we notice that using the *fmincon* method the minimization assigns the same values, for each tenor, to both integrals. This can depend on the fact that they are both

equally unknown and equally relevant for the computation.

The mean root square error $\sqrt{\text{optimaldistance}/(TN)}$ is computed using the two methods and the results we obtain are:

	fmincon	lsqnonlin
MRSE	68.4475	68.4472

Table 5: MRSE in the case of time dependence

We do not notice an improvement from the preceding case, so we can infer that having both parameters time dependent does not give further information.

Furthermore, we notice that for both *fmincon* and *lsqnonlin* the sum $\int_0^t \Sigma_1^2 ds + \int_0^t \Sigma_2^2 ds$ is equal to $\int_0^t \Sigma_1^2 ds$ of point 5, for each tenor.

7 PRICING A DOWN AND IN CALL OPTION

Now we have to price a down and in call option with maturity $T=6$ months with strike $K=500$ and barrier $L=450$ with the models calibrated in points 4, 5 and 6.

To price a down and in call option with the models we previously calibrated we assume to have a daily monitoring and to simulate the underlying dynamics with Monte Carlo.

The dynamic in case of both Σ_1 and Σ_2 constant is:

$$F(t, \tau_1, \tau_2) = F(0, \tau_1, \tau_2) \cdot \exp\left(\int_0^t A(u, \tau_1, \tau_2) du + (\Sigma_1 + \Sigma_2) \int_0^t dW_k(u)\right)$$

In the case in which Σ_1 is time dependent and Σ_2 is constant and equal to 0 the dynamic is:

$$F(t, \tau_1, \tau_2) = F(0, \tau_1, \tau_2) \cdot \exp\left(\int_0^t A(u, \tau_1, \tau_2) du + \int_0^t \Sigma_1(u, \tau_1, \tau_2) dW_k(u)\right)$$

Eventually, in case of both Σ_1 and Σ_2 time dependent the dynamic is the more complete:

$$F(t, \tau_1, \tau_2) = F(0, \tau_1, \tau_2) \cdot \exp\left(\int_0^t A(u, \tau_1, \tau_2) du + \sum_{k=1}^2 \int_0^t \Sigma_k(u, \tau_1, \tau_2) dW_k(u)\right)$$

For computing the dynamic we take into consideration the following mathematical result:

$$\int_0^t f(\tau) dW_\tau \sim N(0, \int_0^t |f(\tau)|^2 d\tau)$$

The results we obtain are summed up in the following table.

	<i>Price</i>	<i>ConfidenceInterval</i>
Both Σ_1 and Σ_2 constant	83.2982	[82.0430 , 84.5534]
Σ_1 time dependent and Σ_2 constant	18.5173	[18.1950 , 18.8397]
Both Σ_1 and Σ_2 time dependent	18.6274	[18.3055 , 18.9493]

Table 6: Prices and their confidence intervals in all 3 cases, daily monitoring

What we notice from the results is that assuming both Σ_1 and Σ_2 constant is very approximative which leads to less reliable price. Furthermore, we notice that by changing the monitoring from daily to weekly or monthly the prices decrease significantly. This is due to the fact that in less monitoring dates there is less probability of touching the barrier. In addition, the case in which both Σ_1 and Σ_2 are constant admits a closed formula:

$$Price = F0 \cdot \exp(-qT)(H/F0)^{2\lambda}N(y) - K \cdot \exp(-rT)(H/F0)^{2\lambda-2}N(y - \sigma\sqrt{T})$$

$$\lambda = \frac{r - q + \sigma^2/2}{\sigma^2}$$

$$y = \frac{\ln(H^2/(F0 \cdot K))}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}$$

The price computed with the closed formula is 42.6245 which is similar to the one obtained in the simulation with monthly monitoring dates (36.9881). As far as the other two cases are concerned, it does not exist a closed formula but we notice that the results obtained are very similar.

There is a significant difference between the first case and the other two because the $\int_0^{T=0.5} \Sigma_1^2 ds$ in the first case is 25% higher compared to other two. In conclusion, we trust the most the price obtained with Σ_1 time dependent and Σ_2 constant. Even though the final price is similar to the case in which both Σ_1 and Σ_2 are time dependent, as well as the confidence interval has the same length, the former has a smaller MRSE when we calibrated in point 5.

References

- [1] Fred Espen Benth, Jurate Saltyte Benth, Steen Koekebakker. *Stochastic Modeling of Electricity and Related Markets*. 2008
- [2] John C. Hull. *Options, Futures, and other Derivatives*. 2009